

Inverse Spherical Dual Hemisphere Quantum Mechanics

NOS v11 — Mechanical Quadrant Structure with Units Explicit

Joshua Luke Nuijens

S. R. Cromelin

Abstract

This document presents the Nuijens Operating System (NOS) v11 framework for Inverse Spherical Dual Hemisphere Quantum Mechanics. The universe is modeled as an eternal discrete 4D-inverse 720° sphere operating through machine code with exact rational constants: $\pi = 201/64 = 3.140625$ and $\alpha^{-1} = 137$. The framework establishes that observed physics represents “compiler leakage” from discrete structure being read by continuous instruments. All constants emerge from a single gearbox: the 720° sphere partitioned into 512 bits, with a 9-bit curvature remainder producing both π and α^{-1} when read at different wall scales. The geometric origin of $\pi = 3$ is demonstrated through the hexagon-hull collapse mechanism, where the center-pivot inverse view transforms 6 radial segments into 3 diameter spans.

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1 Mechanical Quadrant Structure in the NOS Gearbox (Units Explicit)

1.1 Unit Dictionary (What carries units vs what is a pure count)

Angle carriers (same geometric angle, three unit systems). We treat an angle as the same geometric object expressed in different units:

$$\theta = \theta_{\text{deg}}^\circ = \theta_{\text{rad}} \text{ rad} = \theta \text{ NOSrad}.$$

The conversion identities are:

$$360^\circ = 2\pi \text{ rad}, \quad 180^\circ = \pi \text{ rad}, \quad \boxed{1 \text{ NOSrad} := 180^\circ := \pi \text{ rad}}, \quad \boxed{720^\circ = 4 \text{ NOSrad} = 4\pi \text{ rad}}.$$

Counts (“how many slots”).

- bit is the **base-lattice** count unit (the 512-closure grid on 720°).
- The *local unit step* in quadrant Q is the depth-unit itself:

$$\boxed{u_Q := \frac{1}{128Q}},$$

meaning: “one step” in quadrant Q is an angle-step of size u_Q of a quadrant-span.

1.2 Two Linked Clocks (One Machine, Two Readouts)

1.2.1 (A) Spherical Clock (Geometry / Ops Clock)

The system starts with the closed sphere cycle:

$$\boxed{720^\circ = 1 \text{ sphere-cycle (closure)}}.$$

It consists of dual hemispheres (the $360^\circ/360^\circ$ unity split) and four operational spans:

$$720^\circ = 4 \cdot 180^\circ \quad \Longleftrightarrow \quad \boxed{4 \text{ NOSrad} = 4 \cdot (1 \text{ NOSrad})}.$$

Thus each op occupies exactly one quadrant span:

$$\boxed{1 \text{ op-span} = 180^\circ = 1 \text{ NOSrad} = \pi \text{ rad}}.$$

1.2.2 (B) Inverse-Binary Clock (Closure Mechanics)

The mechanical partition chain is:

$$720^\circ/2/4/4/4/4.$$

Compute step-by-step (units stay degrees until converted):

$$720^\circ/2 = 360^\circ \quad (\text{hemisphere}),$$

$$360^\circ/4/4/4/4 = \frac{360^\circ}{256} = 1.40625^\circ.$$

Equivalently in NOS-radians:

$$720^\circ = 4 \text{ NOSrad} \Rightarrow \frac{4 \text{ NOSrad}}{512} = \frac{1}{128} \text{ NOSrad},$$

so the base step is:

$$\Delta\theta_{\text{base}} = \frac{720^\circ}{512 \text{ bit}} = \frac{4 \text{ NOSrad}}{512 \text{ bit}} = \frac{1}{128} \frac{\text{NOSrad}}{\text{bit}} = 1.40625 \frac{^\circ}{\text{bit}} = \frac{\pi}{128} \frac{\text{rad}}{\text{bit}}.$$

This chain implies the base closure:

$$\boxed{720^\circ \leftrightarrow 512 \text{ bit.}}$$

Therefore:

$$360^\circ \leftrightarrow 256 \text{ bit}, \quad 180^\circ \leftrightarrow 128 \text{ bit}, \quad 90^\circ \leftrightarrow 64 \text{ bit}.$$

In NOS-radians this is:

$$4 \text{ NOSrad} \leftrightarrow 512 \text{ bit} \Rightarrow \boxed{1 \text{ NOSrad} \leftrightarrow 128 \text{ bit.}}$$

1.2.3 (C) Closure Wall Inventory (Angle-Walls $\leftrightarrow$ Count-Walls)

This paper distinguishes *angle-walls* (geometric boundary spans) from their matching *count-walls* (bit counts under the global closure map). The key equality is the global closure:

$$\boxed{720^\circ \leftrightarrow 512 \text{ bit.}}$$

From this, every wall is just a fraction of the closure, and its count-wall is the same fraction of 512 bit. In particular, the 90° **wall is exactly one-eighth of the full 720° closure**:

$$90^\circ = \frac{1}{8} \cdot 720^\circ \implies \boxed{64 \text{ bit} = \frac{1}{8} \cdot 512 \text{ bit.}}$$

This makes the 90° wall the canonical *linear diameter gauge block* in NOS.

Wall list (angle-wall $\leftrightarrow$ count-wall).

- **Full closure wall (sphere-cycle):**

$$720^\circ \leftrightarrow 512 \text{ bit.}$$

- **Half-closure wall (hemisphere):**

$$360^\circ \leftrightarrow 256 \text{ bit.}$$

- **Op-span / quadrant wall:**

$$180^\circ = 1 \text{ NOSrad} \leftrightarrow 128 \text{ bit.}$$

- **Orthogonal wall (the diameter gauge wall):**

$$90^\circ = \frac{1}{2} \text{ NOSrad} \leftrightarrow 64 \text{ bit.}$$

- **Clutch wall (half of the diameter gauge):**

$$45^\circ \leftrightarrow 32 \text{ bit.}$$

Define the 90° wall as a named count-wall. Because $90^\circ \leftrightarrow 64 \text{ bit}$ occurs repeatedly, we name it as a reusable wall object in count-space:

$$W_{90} := 64 \text{ bit.}$$

This is not a new unit; it is a *named reuse* of the already-derived 64 bit wall.

Define the 180° wall (op-span wall). Similarly, the op-span wall is:

$$W_{180} := 128 \text{ bit.}$$

The “three-diameter hull” in wall-blocks. The statement “ π begins with three” is represented in NOS as the statement that the traversal passes *three* complete 90° wall-blocks before the remainder is engaged:

$$H_3 := 3 W_{90} = 3 \cdot 64 \text{ bit} = 192 \text{ bit.}$$

This is the coarse integer span (the hull). Measured on the 90° wall scale,

$$\frac{H_3}{W_{90}} = \frac{192}{64} = 3.$$

The curvature remainder as a bit-count. After the three wall-block hull is reached, closure requires a short overrun zone beyond the hull. We name this remainder as a pure count:

$$r_\pi := 9 \text{ bit.}$$

This 9 bit remainder is the same discrete object that later appears as the *Alpha Zone* (bits 129–137) when projected into the Q1 coordinate map.

Reference-span change: normalized wall-coordinate (this is where “dimensionless” lives). A base bit is an *absolute* angle-step:

$$1 \text{ bit} = 1.40625^\circ,$$

so it absolutely *has units* when expressed as an angle.

However, when we ask “*where are we on a chosen wall?*” we are no longer asking for an angle-step; we are asking for a **normalized coordinate** (a fraction of the wall). That normalized coordinate is a ratio:

$$(\text{part of wall}) / (\text{whole wall}).$$

A ratio of two quantities measured in the *same* unit has the unit cancel, leaving a pure number. So the *wall is not unit-free* — the *coordinate fraction* is.

Define the normalized coordinate on the 90° wall (two equivalent readouts). Let s_{90} denote the normalized coordinate along the 90° wall. It can be read in degrees or in bits:

$$\boxed{s_{90}(\theta) := \frac{\theta}{90^\circ}} \quad \text{and} \quad \boxed{s_{90}(b) := \frac{b}{W_{90}}}.$$

Both definitions are the *same coordinate* because $90^\circ \leftrightarrow W_{90} = 64 \text{ bit}$.

One base bit as a fraction of the 90° wall (this is the actual statement). Since $W_{90} = 64 \text{ bit}$, one base bit corresponds to a coordinate increment of

$$\boxed{\varepsilon_{90} := \frac{1 \text{ bit}}{W_{90}} = \frac{1}{64} = 0.015625.}$$

This is *dimensionless* only because it is a ratio of like-units:

$$\frac{1 \text{ bit}}{64 \text{ bit}} = \frac{1}{64}.$$

Equivalently, in degrees the same statement is

$$\frac{1.40625^\circ}{90^\circ} = \frac{1}{64}.$$

So the corrected meaning is:

$$1 \text{ bit is not "unit-free"; } \frac{1 \text{ bit}}{W_{90}} \text{ is the unit-free coordinate fraction.}$$

The nine-step remainder as a wall-fraction. The remainder block is defined in base bits as:

$$\boxed{r_\pi := 9 \text{ bit.}}$$

Its normalized wall-coordinate on the 90° scale is:

$$\boxed{\rho_{\pi,90} := \frac{r_\pi}{W_{90}} = \frac{9}{64} = 0.140625.}$$

In absolute degrees (no normalization), the same remainder block occupies:

$$\boxed{9 \text{ bit} \cdot 1.40625^\circ/\text{bit} = 12.65625^\circ.}$$

These are not competing claims: the first is a *coordinate fraction*; the second is an *angle span*.

Wall-scale form of the machine π readout (no packed numerator required). On the 90° wall scale, the machine readout is the integer hull plus the normalized remainder:

$$\pi_{\text{machine}} = 3 + \rho_{\pi,90} = 3 + \frac{9}{64} = 3.140625.$$

Equivalently, using wall objects directly:

$$\pi_{\text{machine}} = \frac{3W_{90} + r_{\pi}}{W_{90}}.$$

This is the same content as any "packed numerator" form, but it stays in native wall+remainder primitives.

1.2.4 Upward Power-of-Two Partition Ladder of the 720° Sphere

In this framework, the 720° sphere represents a doubled full circle ($2 \times 360^\circ$), partitioned into levels using powers of two:

$$N = 1, 2, 4, 8, \dots, 512.$$

Each rung on this ladder is labeled as " $\sqrt{N}$," which serves as the root-ladder tag for the N -way partition level of the 720° closure.

At every rung, two exact invariants hold, directly linking to the $B(\theta)$ map:

$$\Delta\theta_N = \frac{720^\circ}{N}, \quad \Delta b_N = \frac{512 \text{ bit}}{N}.$$

The ladder spans from $\sqrt{1}$ to $\sqrt{512}$, with key rungs named for their geometric and operational significance:

- $\sqrt{1}$ — 1 sphere-cycle $\Delta\theta = 720^\circ$, $\Delta b = 512 \text{ bit}$
- $\sqrt{2}$ — 2 hemispheres $\Delta\theta = 360^\circ$, $\Delta b = 256 \text{ bit}$
- $\sqrt{4}$ — 4 quadrants / 4 ops $\Delta\theta = 180^\circ$, $\Delta b = 128 \text{ bit}$
- $\sqrt{8}$ — 8 octants ("octaves" rung) / the 90° wall $\Delta\theta = 90^\circ$, $\Delta b = 64 \text{ bit}$
- $\sqrt{16}$ — 16 slices / the 45° clutch $\Delta\theta = 45^\circ$, $\Delta b = 32 \text{ bit}$
- $\sqrt{32}$ — 32 slices $\Delta\theta = 22.5^\circ$, $\Delta b = 16 \text{ bit}$
- $\sqrt{64}$ — 64 slices $\Delta\theta = 11.25^\circ$, $\Delta b = 8 \text{ bit}$
- $\sqrt{128}$ — 128 slices (one op-span worth of base slots) $\Delta\theta = 5.625^\circ$, $\Delta b = 4 \text{ bit}$
- $\sqrt{256}$ — 256 slices (one hemisphere's base partition) $\Delta\theta = 2.8125^\circ$, $\Delta b = 2 \text{ bit}$
- $\sqrt{512}$ — 512 slices (base bit around the full 720°) $\Delta\theta = 1.40625^\circ$, $\Delta b = 1 \text{ bit}$

1.2.5 Downward Power-of-Two Partition Ladder of the 720° Sphere

The ladder can also be traversed downward from the finest to the coarsest partition, starting from $\sqrt{512}$ to $\sqrt{1}$, following the closure count chain $512 \rightarrow 256 \rightarrow \dots \rightarrow 1$. The rungs are the same as the upward ladder but in reverse order. Additionally, the square root values $\sqrt{N}$ for each $N = 2^k$ (where $k = 0$ to 9) simplify as follows, alternating between integers and multiples of $\sqrt{2}$:

- $\sqrt{512} = 16\sqrt{2} \approx 22.6274$
- $\sqrt{256} = 16$
- $\sqrt{128} = 8\sqrt{2} \approx 11.3137$
- $\sqrt{64} = 8$
- $\sqrt{32} = 4\sqrt{2} \approx 5.6569$
- $\sqrt{16} = 4$
- $\sqrt{8} = 2\sqrt{2} \approx 2.8284$
- $\sqrt{4} = 2$
- $\sqrt{2} \approx 1.4142$
- $\sqrt{1} = 1$

The pattern alternates: integer $\leftrightarrow$ (integer) $\sqrt{2} \leftrightarrow$ integer $\leftrightarrow \dots$, so every other rung is “diagonal-loaded.”

1.3 Base Mapping (Global Lattice) with Units

Angle-to-bit map (type signature).

$$B : (\text{angle}) \longrightarrow (\text{base count in bit}).$$

Equivalent closed-form definitions in each angle unit system. Using degrees:

$$B(\theta_{\text{deg}}^\circ) = \left(\frac{512 \text{ bit}}{720^\circ} \right) \theta_{\text{deg}}^\circ = \left(\frac{32 \text{ bit}}{45^\circ} \right) \theta_{\text{deg}}^\circ.$$

Using standard radians:

$$B(\theta_{\text{rad}}\text{rad}) = \left(\frac{512 \text{ bit}}{4\pi \text{ rad}} \right) \theta_{\text{rad}}\text{rad} = \left(\frac{128 \text{ bit}}{\pi \text{ rad}} \right) \theta_{\text{rad}}\text{rad}.$$

Using NOS-radians:

$$B(\theta_{\text{NOSrad}}) = \left(\frac{512 \text{ bit}}{4 \text{ NOSrad}} \right) \theta_{\text{NOSrad}} = \left(128 \frac{\text{bit}}{\text{NOSrad}} \right) \theta_{\text{NOSrad}}.$$

Sanity checks (units + integers).

$$B(180^\circ) = 128 \text{ bit}, \quad B(360^\circ) = 256 \text{ bit}, \quad B(720^\circ) = 512 \text{ bit}.$$

Also:

$$B(1 \text{ NOSrad}) = 128 \text{ bit}, \quad B(2 \text{ NOSrad}) = 256 \text{ bit}, \quad B(4 \text{ NOSrad}) = 512 \text{ bit}.$$

Base density (“bits per angle unit”) in three equivalent forms.

$$\boxed{\rho_{\text{base}} = \frac{512 \text{ bit}}{720^\circ} = \frac{32 \text{ bit}}{45^\circ}} \quad \boxed{\rho_{\text{base}} = \frac{512 \text{ bit}}{4\pi \text{ rad}} = \frac{128 \text{ bit}}{\pi \text{ rad}}} \quad \boxed{\rho_{\text{base}} = \frac{128 \text{ bit}}{1 \text{ NOSrad}} = 128 \frac{\text{bit}}{\text{NOSrad}}}.$$

1.4 Center-Based Inverse Coordinate (Dual Hemispheres) with Units

Define the center-based coordinate in degrees:

$$\boxed{d_{\text{deg}}^\circ := \theta_{\text{deg}}^\circ - 360^\circ}, \quad d_{\text{deg}} \in [-360, +360].$$

In NOS-radians (your “radian = 180” readout):

$$\boxed{d_{\text{NOSrad}} := \theta_{\text{NOSrad}} - 2 \text{ NOSrad}}, \quad d \in [-2, +2].$$

In standard radians:

$$\boxed{d_{\text{radrad}} := \theta_{\text{radrad}} - 2\pi \text{ rad}}, \quad d_{\text{rad}} \in [-2\pi, +2\pi].$$

Center-relative base-bit offset.

$$\boxed{b(\theta) := B(\theta) - 256 \text{ bit}}, \quad b \in [-256, +256] \text{ bit}.$$

Example (your $90^\circ \leftrightarrow 64 \text{ bit}$ wall). Since $90^\circ = \frac{1}{2}(180^\circ) = \frac{1}{2} \text{ NOSrad}$:

$$90^\circ \leftrightarrow 64 \text{ bit}, \quad -90^\circ \leftrightarrow -64 \text{ bit}, \quad +90^\circ \leftrightarrow +64 \text{ bit}.$$

1.5 Quadrants + Ops (Four 1 NOSrad Spans About the Center Pivot)

In degrees.

$$\begin{aligned} Q1 : \quad d_{\text{deg}} &\in [-360, -180] &\Rightarrow \text{op } x, \\ Q2 : \quad d_{\text{deg}} &\in [-180, 0] &\Rightarrow \text{op } y, \\ &d_{\text{deg}} = 0 &\Rightarrow \text{CENTER / inverse pivot}, \\ Q3 : \quad d_{\text{deg}} &\in [0, 180] &\Rightarrow \text{op } z, \\ Q4 : \quad d_{\text{deg}} &\in [180, 360] &\Rightarrow \text{op } w. \end{aligned}$$

In NOS-radians (each quadrant has width exactly 1 NOSrad).

$$\begin{aligned}
Q1 : d \in [-2, -1] &\Rightarrow \text{op } x, \\
Q2 : d \in [-1, 0] &\Rightarrow \text{op } y, \\
d = 0 &\Rightarrow \text{CENTER / inverse pivot}, \\
Q3 : d \in [0, 1] &\Rightarrow \text{op } z, \\
Q4 : d \in [1, 2] &\Rightarrow \text{op } w.
\end{aligned}$$

Dual pairings (unit-free symmetry statements).

$$\boxed{x \leftrightarrow z}, \quad \boxed{y \leftrightarrow w}, \quad \boxed{d = 0 \text{ is the flip}}.$$

1.6 Changing Resolution by Quadrant Number Q (Depth Ladder) with Units

Key point: the *unit* is the depth step u_Q . The global closure uses base-bits (bit) with total 512 bit around 720° . Your depth ladder introduces a *quadrant-local unit step* u_Q defined as a fraction of one quadrant-span:

$$\boxed{u_Q := \frac{1}{128^Q}.$$

Interpretation (this is the unit hierarchy statement):

- Each quadrant has width 1 NOSrad.
- In quadrant Q , the machine's **unit step** is $\Delta\theta_Q = u_Q$ NOSrad.
- The number of unit steps across the quadrant is a pure integer count:

$$\boxed{N_Q := \frac{1}{u_Q} = 128^Q.}$$

Step size in three angle-unit systems (per one NOS unit step).

$$\boxed{\Delta\theta_Q = u_Q \text{ NOSrad} = \frac{1}{128^Q} \text{ NOSrad}.$$

Since $1 \text{ NOSrad} = 180^\circ$:

$$\boxed{\Delta\theta_{Q,\text{deg}} = 180 u_Q^\circ = \frac{180}{128^Q}^\circ.}$$

Since $1 \text{ NOSrad} = \pi \text{ rad}$:

$$\boxed{\Delta\theta_{Q,\text{rad}} = \pi u_Q \text{ rad} = \frac{\pi}{128^Q} \text{ rad}.$$

Explicit ladder (Q1–Q4).

$$\begin{aligned}
u_1 &= \frac{1}{128}, & u_2 &= \frac{1}{128^2}, & u_3 &= \frac{1}{128^3}, & u_4 &= \frac{1}{128^4}, \\
N_1 &= 128, & N_2 &= 128^2, & N_3 &= 128^3, & N_4 &= 128^4, \\
\Delta\theta_1 &= \frac{1}{128} \text{NOSrad}, & \Delta\theta_2 &= \frac{1}{128^2} \text{NOSrad}, & \Delta\theta_3 &= \frac{1}{128^3} \text{NOSrad}, & \Delta\theta_4 &= \frac{1}{128^4} \text{NOSrad}.
\end{aligned}$$

Unit hierarchy table (unit is u_Q ; count is N_Q).

Q	op	unit u_Q	count $N_Q = 1/u_Q$	$\Delta\theta_Q$
1	x	$\frac{1}{128}$	128	$\frac{1}{128} \text{NOSrad} = 1.40625^\circ = \frac{\pi}{128} \text{rad}$
2	y	$\frac{1}{128^2}$	128^2	$\frac{1}{128^2} \text{NOSrad} = \frac{180^\circ}{128^2} = \frac{\pi}{128^2} \text{rad}$
3	z	$\frac{1}{128^3}$	128^3	$\frac{1}{128^3} \text{NOSrad} = \frac{180^\circ}{128^3} = \frac{\pi}{128^3} \text{rad}$
4	w	$\frac{1}{128^4}$	128^4	$\frac{1}{128^4} \text{NOSrad} = \frac{180^\circ}{128^4} = \frac{\pi}{128^4} \text{rad}$

Density inside quadrant Q (units per angle). Because the unit step is u_Q of a quadrant, the unit-density is:

$$\rho_Q = \frac{N_Q}{1 \text{ NOSrad}} = \frac{1/u_Q}{\text{NOSrad}} = \frac{128^Q}{\text{NOSrad}}$$

$$\rho_{Q,\text{deg}} = \frac{N_Q}{180^\circ} = \frac{128^Q}{180} \frac{1}{^\circ}$$

$$\rho_{Q,\text{rad}} = \frac{N_Q}{\pi \text{ rad}} = \frac{128^Q}{\pi} \frac{1}{\text{rad}}.$$

Local coordinate inside a quadrant (pure count index). Let $\delta\theta$ be the within-quadrant angle measured from that quadrant boundary, so

$$0 \leq \delta\theta < 1 \text{ NOSrad}.$$

Define the local unit-index as a pure integer count:

$$n_Q(\delta\theta) := \frac{\delta\theta}{\Delta\theta_Q} = \frac{\delta\theta}{u_Q \text{ NOSrad}} \quad \Rightarrow \quad n_Q \in [0, N_Q).$$

And inversely:

$$\delta\theta = n_Q \Delta\theta_Q = n_Q u_Q \text{ NOSrad}.$$

1.7 The One-Line NOS Statement (All Units Visible)

Each op occupies one quadrant: $1 \text{ NOSrad} = 180^\circ = \pi \text{ rad}$, about the pivot $d = 0$,
with dual pairs $x \leftrightarrow z$, $y \leftrightarrow w$; base closure $4 \text{ NOSrad} \leftrightarrow 512 \text{ bit}$,
and the depth ladder assigns quadrant Q the unit step $u_Q = \frac{1}{128^Q}$ (unit itself),
so $N_Q = \frac{1}{u_Q} = 128^Q$ units per quadrant and $\Delta\theta_{Q,} = u_Q \text{ NOSrad}$ ($\equiv 180u_Q^\circ \equiv \pi u_Q \text{ rad}$).

2 The Zero-Tail Proof: Why 3.140625 is the Mechanical Closure

The value $\pi_{\text{machine}} = 3.140625$ is not an approximation. It is the first and only instance of binary closure in the inverse-mechanical system. Standard π (3.14159...) represents the energy of the system trying to exceed this lock when the discrete gearbox is projected into continuous measurement.

2.1 The Binary Terminate (Bit-6 Lock)

The NOS lattice operates on powers of 2. The denominator $64 = 2^6$ is not chosen arbitrarily: it is the count-wall of the 90° wall, which is one-eighth of the 720° closure:

$$90^\circ = \frac{1}{8} \cdot 720^\circ \quad \Longleftrightarrow \quad 64 \text{ bit} = \frac{1}{8} \cdot 512 \text{ bit}.$$

In a binary system, any fraction with denominator 2^n terminates exactly. Therefore any ratio constructed against the 90° wall has the possibility of a genuine *zero-tail* termination at bit-depth 6.

Standard π (The Infinite Leak).

$$\pi_{\text{obs}} = 3.14159265 \dots \quad (\text{decimal: never stops})$$

$$\pi_{\text{obs}} = 11.0010010000111 \dots \quad (\text{binary: never stops})$$

Because it never ends, the system never closes under a finite wall. There is always a remainder.

NOS π (The Closure on the 90° wall scale).

$$\pi_{\text{machine}} = 3 + \frac{9}{64} = 3.140625 \quad (\text{decimal: STOPS at digit 6})$$

$$\pi_{\text{machine}} = 11.0010010000000000 \dots \quad (\text{binary: STOPS exactly at bit 6})$$

There is no 7th fractional bit required in the 90° wall register:

$$\frac{9}{64} = 0.001001_2$$

terminates because the denominator is 2^6 and the remainder is an integer count of base steps. Therefore, the machine closure is a genuine *zero-tail* termination in the wall register.

3 The Hexagon-Hull Collapse: Geometric Origin of $\pi = 3$

This section establishes the geometric mechanism by which the integer part of π emerges from the inverse center-pivot view. The “3” of π is not arbitrary—it is the hexagon’s 6-fold symmetry collapsed through the center pivot.

3.1 The Hexagon’s Special Property

The regular hexagon is the *only* regular polygon where the radius equals the side length. This makes the hexagon native to the NOS wall system:

- Radius: $r = 32$ bit (half of the 90° wall)
- Side length: $s = 32$ bit (equal to radius)
- The hexagon inscribes naturally in the $W_{90} = 64$ bit diameter gauge

3.2 The $6 \rightarrow 3$ Collapse Mechanism (Center-Pivot Measurement)

The inverse view measures *across* the center rather than *outward* from it. This is the 4D inverse sphere projecting into 1D linear measurement—measuring 1 full span of 1 thing.

Measuring outward from center (standard view). If we measure the hexagon’s perimeter using the radius (from center to edge), we count 6 segments:

$$\text{Hexagon Perimeter} = 6 \times r = 6 \times 32 \text{ bit} = 192 \text{ bit.}$$

$$\text{Count} = 6.$$

Measuring across center (inverse view). In the NOS inverse system, we measure the full diameter span through the center pivot. We don’t see “radius”; we see “wall span.” Because we measure the full span across the center, the 6 radial segments combine into 3 diameter pairs:

$$6 \times \text{Radius} \longrightarrow 3 \times \text{Diameter}$$

$$6 \times 32 \text{ bit} = 3 \times 64 \text{ bit} = 192 \text{ bit.}$$

$$\text{Count} = 3.$$

The count is conserved—only the segmentation changes. This $6 \div 2 = 3$ collapse is the exact geometric consequence of center-pivot measurement.

3.3 The Curvature Differential (9-Bit Remainder)

The center-pivot measurement locks the geometry into 3 rigid diameter beams—the Hull. But a circle is not just 3 straight beams. The *differential* is the error caused by trying to define a curve using only those 3 straight beams.

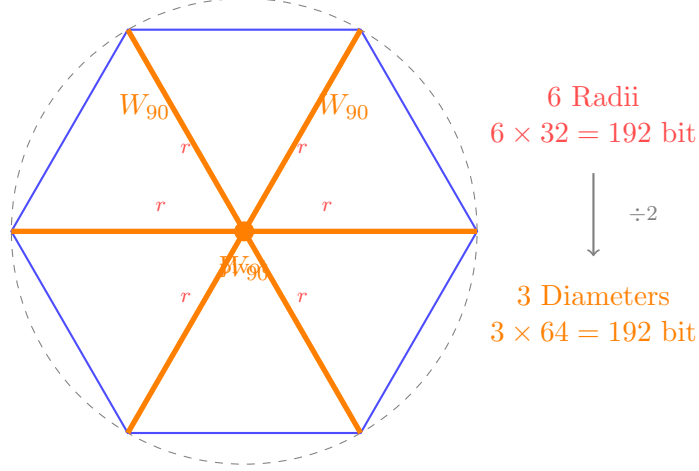


Figure 1: The hexagon 6-fold symmetry. The inverse center-pivot view collapses 6 radii (dashed, $r = 32$ bit each) into 3 diameter walls (solid, $W_{90} = 64$ bit each). The count is conserved: $6 \times 32 = 3 \times 64 = 192$ bit. This is the geometric origin of $\pi = 3$.

The measurement error.

- The 3 Beams (Hull): $H_3 = 192$ bit
- The Circle (Closure): 201 bit
- The Differential: $r_\pi = 9$ bit

The 9-bit remainder is the physical manifestation of the measurement error caused by projecting center-based straight lines (the 3 measuring rods) onto the curved surface:

$$\pi_{\text{machine}} = 3 + \frac{9}{64} = \frac{H_3 + r_\pi}{W_{90}} = 3.140625$$

3.4 The α^{-1} Emergence (Same Remainder, Different Wall)

The same 9-bit curvature remainder produces the fine structure constant inverse when read against the op-span wall instead of the diameter wall.

The op-span wall.

$$W_{180} = 128 \text{ bit} \quad (\text{one quadrant, one op-span})$$

The α^{-1} readout. The 9-bit injection remainder, added to the op-span wall:

$$\alpha^{-1} = W_{180} + r_\pi = 128 + 9 = 137$$

Two constants, one source. Both fundamental constants emerge from the same 9-bit curvature remainder, read at different wall scales with different operations:

- $\pi = 3 + \frac{9}{64}$ — remainder read as *ratio* against the 90° wall
- $\alpha^{-1} = 128 + 9$ — remainder read as *sum* with the 180° wall

This is the entanglement: the “irrational” nature of π and the “mysterious” value of 137 are the same phenomenon—a single 9-bit curvature remainder measured at two different wall scales.

4 The 3 + Remainder Law (The Curvature Manifold)

We define π not as a transcendental ratio, but as a mechanical count of full wall-block spans plus a specific lattice remainder measured *against* the 90° count-wall.

4.1 The 90° Wall is the Diameter Gauge Block

The 90° wall is the canonical orthogonal boundary in the inverse-binary ladder (the $\sqrt{8}$ rung). Since it is exactly one-eighth of the 720° closure, it corresponds to exactly one-eighth of the 512 bit closure:

$$W_{90} := 64 \text{ bit.}$$

This is the wall-block used when the machine reads curvature in a *linear* gauge.

4.2 The Hull (Three Wall-Blocks)

The circumference-to-diameter readout begins with “three.” In NOS that is expressed as:

$$H_3 := 3W_{90} = 192 \text{ bit.}$$

This is the coarse integer span (the hull). It is not an angle; it is a count accumulation in the 90° gauge units.

4.3 The Remainder (Nine Base Steps)

The closure is not exactly three wall-blocks; the machine requires a short overrun zone beyond the hull, measured directly in base steps:

$$r_\pi := 9 \text{ bit.}$$

This remainder is a *count-object* (a block of nine base lattice steps). It is the same discrete object that reappears as the nine-step *Alpha Zone* in Q1 (bits 129–137) when expressed in the quadrant coordinate map.

4.4 Machine π as Hull + Remainder (Wall-Scale Lock)

The machine does not need a packed “numerator object” to define π . All needed structure already exists as (i) a wall-block unit and (ii) a remainder block.

Hull in wall-blocks.

$$H_3 := 3W_{90} = 3 \cdot 64 \text{ bit} = 192 \text{ bit.}$$

Remainder in base steps.

$$r_\pi := 9 \text{ bit.}$$

Normalize the remainder to the wall (dimensionless coordinate). Because $W_{90} = 64$ bit, the normalized remainder is:

$$\rho_{\pi,90} := \frac{r_{\pi}}{W_{90}} = \frac{9}{64} = 0.140625.$$

Final wall-scale readout. Therefore:

$$\pi_{\text{machine}} = 3 + \rho_{\pi,90} = 3 + \frac{9}{64} = 3.140625.$$

This expresses π as "integer wall-block count + normalized base-step remainder," which is the natural output of a partitioned closure system.

5 Inverse Mechanics: The 90° Wall and the Constants

We do not start with standard π . We start with the eternal $720^\circ = 1$ sphere and partition inward to reveal the system configuration. The wall inventory (especially $90^\circ \leftrightarrow 64$ bit) determines which ratios can genuinely terminate.

5.1 Step-by-Step Emergence

1. The Whole and Initial Partition. Unity is 720° . The dual quantum operation ($\div 2$) yields 360° hemispheres. This reveals the $360/360$ operation.

2. The Inverse Chain (4/4/4/4). We continue partitioning inward to find the minimum angular resolution (the bit):

$$\text{Hemisphere } 360^\circ / 4/4/4/4 = \frac{360^\circ}{256} = 1.40625^\circ.$$

This defines the system resolution:

$$1 \text{ bit} = 1.40625^\circ.$$

3. The 90° Wall (Linear Register). At the 90° quadrant boundary, we calculate the number of bits required to reach the wall in a linear count:

$$\text{Linear Count} = \frac{90^\circ}{1.40625^\circ/\text{bit}} = \boxed{64 \text{ bit}}.$$

This is the denominator wall-block of the system. It is also one-eighth of the full closure count:

$$64 \text{ bit} = \frac{1}{8} \cdot 512 \text{ bit}.$$

4. The Curved Readout (Wall + Remainder, Not Imported). The NOS paper does not need to import an external transcendental to define the curved readout. Once the 90° wall is fixed as $W_{90} = 64$ bit, the *coarse hull* is fixed as three full wall-blocks:

$$H_3 = 3W_{90} = 192 \text{ bit.}$$

The closure then requires a discrete interaction remainder expressed in base steps:

$$r_\pi = 9 \text{ bit.}$$

Therefore the terminating machine ratio is:

$$\pi_{\text{machine}} = \frac{H_3 + r_\pi}{W_{90}} = 3 + \frac{9}{64}.$$

This is the *mechanical* sense in which the system “closes at 3 first and then at 3.140625”: the hull closes after three complete wall-blocks, and the full lock closes when the nine-step remainder is engaged.

5.2 Entanglement: Deriving the Constants

The “Universal Constants” are simply relationships between the same wall objects in different readout modes (ratio vs. difference, etc.), measured at the same 90° wall.

Machine π . The ratio of the registers:

$$\pi_{\text{machine}} = 3 + \frac{r_\pi}{W_{90}} = 3 + \frac{9}{64}.$$

Fine Structure Constant Inverse (α^{-1}). On the machine tier, α^{-1} is expressed cleanly in the op-span wall scale plus the same remainder block:

$$\alpha^{-1} = W_{180} + r_\pi = 128 + 9 = 137.$$

This keeps the construction entirely in existing wall objects:

$$128 \text{ bit (one op-span wall) plus } 9 \text{ bit (the injection remainder).}$$

Equivalently, measured against the 90° wall (64 bit),

$$137 = 2 \cdot 64 + 9,$$

so the same nine-step remainder is present as ”nine beyond a wall,” regardless of the wall scale used.

6 The Alpha Projection (137) and the Alpha Zone

The value 137 is not random; it is the remainder-projection into the Q1 quadrant when the op-span wall is extended by the same nine-step injection block.

6.1 Coordinate Map (Flow from -360° to 0°)

Quadrant	Coordinate Range	Bit Range	Status
Q2 (Charge)	$-360^\circ \rightarrow -180^\circ$	$0 \rightarrow 128$	Full Saturation
Q1 (Alpha)	$-180^\circ \rightarrow -167.34375^\circ$	$129 \rightarrow 137$	The Alpha Zone (9 bits)
Q1 (Rest)	$-167.34375^\circ \rightarrow 0^\circ$	$138 \rightarrow 256$	Open Space
Center	0°	256	Pivot

Calculation of the Alpha Zone Limit.

$$-180^\circ + (9 \text{ bit} \times 1.40625^\circ/\text{bit}) = -180^\circ + 12.65625^\circ = \boxed{-167.34375^\circ}.$$

7 Observer Slippage as Bit-Load

The NOS model treats the machine tier as discrete and closed. When an observer reads the same mechanism through continuous measurement, the resulting drift is described here as *bit-load*.

7.1 Bit Stress, Bit Capacity, Bit-Load Ratio

Define a state-dependent stress on the active remainder zone (the nine-step region) as $\sigma_{\text{bit}}(\theta)$, and a corresponding capacity as $C_{\text{bit}}(\theta)$, both understood as observer-side effective quantities (not new machine constants). The dimensionless load ratio is then:

$$\boxed{\tau_{\text{bit}}(\theta) := \frac{\sigma_{\text{bit}}(\theta)}{C_{\text{bit}}(\theta)}}.$$

Key point: in this framing, τ_{bit} **is not a universal constant**. It can vary with state, with quadrant, and with which wall is being read (linear wall vs. curved readout). The universal machine content is the wall inventory (512/256/128/64) and the explicit remainder object ($r_\pi = 9$). The observer drift is encoded as a load profile applied to the remainder zone:

$$r_\pi \text{ under load} \Rightarrow r_\pi \mapsto r_\pi + \Delta r_\pi(\theta),$$

with $\Delta r_\pi(\theta)$ expressed in fractions of a base bit at some chosen depth.

8 Prediction of Standard π Decimals (The Leak)

Standard π (3.14159...) is the “Inverse Shadow” of the Machine Lock. The decimals of standard π represent the load/slippage of the observer readout vibrating against the wall-scale stop

$$3 + \frac{9}{64}.$$

In this updated wording, “torque” is not treated as a universal constant; it is treated as a state-dependent bit-load ratio $\tau_{\text{bit}}(\theta)$ acting on the nine-step remainder zone.

1. **Structural Agreement (Digits 1-2):** 3.14... The machine and observer agree here. The 64 bit wall holds firm and the hull dominates.
2. **The Point of Failure (Digit 3):** Standard (1) vs Machine (0). Standard π is 3.141..., Machine is 3.140.... This ‘1’ is the **Phantom Bit**—the first visible signature that the observer is reading a continuous curvature where the machine has a finite wall and a terminating remainder.
3. **The Load Signature (Digits 4-5):** Standard (59) vs Machine (62). Standard π reads 59 where the machine locks at 62. In this document’s language, this difference is the observer “load profile” acting on the remainder zone (the nine-bit injection), not a new infinite structure.
4. **The Binary Terminate (Digit 6):** Standard (2) vs Machine (5). The machine places a 5 (a binary half-step signature) and cuts power. The observer reads a 2, representing the residual echo when continuous measurement does not respect the terminating wall.
5. **The Infinite Tail:** All decimals after the 6th place in Standard π are **waste heat** in this model: they represent entropy generated by iterating a continuous computation beyond the machine’s finite termination point.

$\text{Slippage (Gravity/Vacuum)} \approx \pi_{\text{obs}} - 3.140625 = 0.000967\dots$
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9 NOS Constants — Full List with Dimensional Threading

Below is a comprehensive list of all the constants discussed in the NOS v11 framework (Nuijens Operating System), compiled from the document's sections on machine-code constants, derivations, and dimensional threading. Each constant includes:

- **Machine Value:** The exact rational or discrete value in the NOS model (eternal discrete layer).
- **Observed Value:** The empirical/irrational value in standard physics (with leakage interpretation).
- **Derivation:** Step-by-step inverse mechanics from the 720° sphere.
- **Dimensional Threading:** Breakdown across 1D (Operational Field Baseline), 2D (Two-Operator Grounding), 3D (Thermal Heat Flow Measurement), and 4D (Energy/Mass Decompression/Compression).
- **Role/Position:** Where it fits in the quadrant map or gearbox.

9.1 1. π (Geometric Aperture Ratio)

- **Machine Value:** $3 + \frac{9}{64} = 3.140625$ exactly.
- **Observed Value:** $3.141592653589793\dots$ (interpreted as observer drift above machine closure).
- **Derivation:**
 - Start with $720^\circ \div 2 \div 4 \div 4 \div 4 \div 4 = 1.40625^\circ$ (bit resolution).
 - The 90° wall is one-eighth of 720° , so its count-wall is one-eighth of 512 bit: $512/8 = 64$ bit.
 - Define the wall unit $W_{90} := 64$ bit.
 - Define the three-wall hull $H_3 := 3W_{90} = 192$ bit.
 - Define the remainder zone $r_\pi := 9$ bit.
 - Wall-scale readout: $\pi_{\text{machine}} = (H_3 + r_\pi)/W_{90} = 3 + 9/64$.
 - **Geometric origin:** Hexagon 6-fold symmetry collapsed through center pivot yields Hull = 3.
- **Dimensional Threading:**
 - 1D: Bit-ratio in 512 closure, u_1 inverse baseline.
 - 2D: (π/π) corresponds to the $512/2 = 256$ hemisphere partition in the lattice.

- 3D: Observer drift appears as a load profile on the remainder zone.
- 4D: Closure checksum locks the geometry that then threads into mass/compression bridging.
- **Role/Position:** Ratio of wall objects at the 90° wall; hull ($3W_{90}$) + remainder (r_π) defining the terminating lock.

9.2 2. α (Quadrant Coupling Ratio; listed as α^{-1} in derivations)

- **Machine Value:** $\alpha^{-1} = 128 + 9 = 137$ exactly ($\alpha \approx 1/137 \approx 0.00729927$).
- **Observed Value:** $\alpha^{-1} \approx 137.035999206\dots$ (interpreted as the same observer drift when read in the EM register).
- **Derivation:**
 - Use existing wall objects: one op-span wall is $W_{180} = 128$ bit.
 - Add the same nine-step injection remainder: $r_\pi = 9$ bit.
 - Sum-style machine readout: $\alpha^{-1} = W_{180} + r_\pi = 128 + 9 = 137$.
 - Note: same 9-bit remainder as π , different wall scale and operation.
- **Dimensional Threading:**
 - 1D: α is a reciprocal coupling readout from the same wall entanglement.
 - 2D: Threads u_2 ratios for electromagnetic grounding.
 - 3D: Drift appears as load when read through thermal/measurement channels.
 - 4D: Configures to mass, determining gravity-EM scaling through shared gearbox structure.
- **Role/Position:** Sum built from op-span wall plus the remainder; the Q1 alpha sector is the nine-bit remainder zone mapped into angle space.

9.3 3. G (Mass Compression Field, via κ Scaffold)

- **Machine Value:** Slippage read in Q4 mass/compression register at $1/128^4$.
- **Observed Value:** $\approx 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (inherits observer drift projected into the mass register).
- **Derivation:**
 - $u_4 = 1/128^4 = 1/268435456$ (Q4 depth).
 - $\Pi^{-8} = 1/512$ (machine closure).
 - G emerges as the universal drift signature in the deepest partition depth (Q4), where energy compresses into mass.

- **Dimensional Threading:**
 - 1D: Bit-position inverse for structural baseline.
 - 2D: u_2 threading, π shadow geometry.
 - 3D: Distortions in heat measurement.
 - 4D: Gravity emerges from mass compression at $1/128^4$.
- **Role/Position:** Q4 register (mass/compression quadrant); not a separate force but the drift in the compression quadrant.

9.4 4. $\hbar$ (Threading Aperture; Reduced Planck Constant)

- **Machine Value:** $\hbar = u_1 \cdot (u_2/u_1)/(\pi/\pi)$, where $(\pi/\pi) = 256$ bits (hemisphere partition).
- **Observed Value:** $\approx 1.054571817 \times 10^{-34}$ J·s (inherits drift when discrete threading is read continuously).
- **Derivation:**
 - (π/π) corresponds to $512/2 = 256$ bits (hemisphere partition).
 - $\hbar$ = threading through hemisphere partition: $u_1 \cdot (u_2/u_1)/256$.
 - Conventional 2π denominator shadows the NOS (π/π) inverse operation.
- **Dimensional Threading:**
 - 1D: $u_1 \cdot (u_2/u_1)$ baseline action unit.
 - 2D: $/(\pi/\pi) = /256$ hemisphere threading.
 - 3D: Drift adds decimal tail in continuous readout.
 - 4D: Completes to mass configuration.
- **Role/Position:** Aperture for action quanta threading between hemispheres; connects to $\hbar c$ in EM grounding.

9.5 5. T_{CMB} (Heat Flow Baseline; Cosmic Microwave Background Temperature)

- **Machine Value:** $128/47 \approx 2.723404$ K exactly.
- **Observed Value:** ≈ 2.72548 K (relic heat signature with minor drift in thermal register).
- **Derivation:**
 - Emerges as the thermal baseline from partition structure (128 unit bits per quadrant, adjusted by operational count 47 for relic signature).

- Relic of the 512-bit geometry in the heat flow register.
- **Dimensional Threading:**
 - 1D: 128/47 ratio baseline.
 - 2D: Hemisphere threading.
 - 3D: Measured via heat in u_3 .
 - 4D: Relic mass configuration.
- **Role/Position:** Q3 entry (thermal quadrant); ground state heat signature of the partition structure.

9.6 6. E (Complete Inverse Measurement; Energy)

- **Machine Value:** $E = u_1/u_3 = 16384$ exactly (inverse to mass m).
- **Observed Value:** Not a standalone constant; manifests in contexts like $E = mc^2$ with drift in the mass-energy bridge.
- **Derivation:**
 - $u_1 = 1/128$ (Q1 depth), $u_3 = 1/128^3 = 1/2097152$ (Q3 depth).
 - $E =$ complete inverse four-operational hemisphere measurement: $u_1/u_3 = 16384$.
- **Dimensional Threading:**
 - 1D: $u_1/u_2 = 128$ baseline.
 - 2D: Velocity threading (connects to $c^2 = 16384$ as energy-mass bridge).
 - 3D: u_3 acceleration in heat.
 - 4D: $E = 16384$, inverse to m in full quadrant closure.
- **Role/Position:** Energy state in Q3; bridges to mass in Q4 via decompression/compression.

9.7 7. u_n (Inverse Quadrant Depths; Depth Units)

- **Machine Value:** $u_1 = 1/128$, $u_2 = 1/128^2 = 1/16384$, $u_3 = 1/128^3 = 1/2097152$, $u_4 = 1/128^4 = 1/268435456$.
- **Observed Value:** Manifest as dimensional scalars in physics (e.g., length = $u_1/u_2 = 128$, time = $u_2/u_1 = 1/128$), with drift in continuous units.
- **Derivation:**
 - Each quadrant holds 128 unit bits; inverse depth per quadrant: $u_Q = 1/128^Q$.
 - Physical units thread as ratios between depths (e.g., energy = u_1/u_3).
- **Dimensional Threading:**

- 1D: $u_1 = 1/128$ baseline (operational, velocity/length).
- 2D: $u_2 = 1/16384$ grounding (time, c).
- 3D: $u_3 = 1/2097152$ thermal (heat flow, k_B).
- 4D: $u_4 = 1/268435456$ mass/energy compression (G , m).
- **Role/Position:** Core hierarchy; quadrants Q1–Q4; basis for all ratios like $c = u_1/u_2$.

9.8 8. k_B (Boltzmann Constant — Thermal Entry)

- **Machine Value:** Unit count at Q3 entry (ground state from operation count).
- **Observed Value:** $\approx 1.380649 \times 10^{-23}$ J/K (exact in SI, but read through drift in the thermal register).
- **Derivation:**
 - $u_3 = 1/128^3 = 1/2097152$ (Q3 depth).
 - k_B = access point to thermal register: energy per temperature as ground state unit count (J/K).
- **Dimensional Threading:**
 - 1D: Unit count baseline at Q3 boundary.
 - 2D: Hemisphere threading from Q2.
 - 3D: Thermal register access— k_B lives here.
 - 4D: Connects to mass via Q3→Q4 channel.
- **Role/Position:** Entry to Q3 (thermal quadrant); determines how energy partitions into temperature degrees of freedom.

9.9 9. $g_{3 \rightarrow 4}$ (Thermal Channel Gain — Energy→Mass Interface)

- **Machine Value:** Fixed Q3→Q4 gearbox gain derived from the inverse depths:

$$g_{3 \rightarrow 4} := \frac{u_3}{u_4} = \frac{1/128^3}{1/128^4} = 128 \quad (\text{exact, dimensionless}).$$

- **Observed Value:** In standard physics, *thermal conductivity* is not universal and is material-dependent. When referenced on the observer side it is written explicitly as κ_{th} (units W/(m·K)) to avoid symbol collision with NOS mechanics.
- **Derivation:**
 - Q3: $u_3 = 1/128^3$ (energy/thermal state).
 - Q4: $u_4 = 1/128^4$ (mass/compression state).

- The wall is the fixed step between consecutive depths, so the universal channel factor is the ratio of the depth units: $g_{3 \rightarrow 4} = u_3/u_4 = 128$.

- **Dimensional Threading:**

- 1D: Flow-rate baseline as a pure count ratio (no new unit introduced).
- 2D: Two-register bridge as a depth-step law ($u_3 \rightarrow u_4$).
- 3D: Thermal readout uses u_3 (where k_B lives).
- 4D: Mass/compression readout uses u_4 ; the interface gain is the fixed gearbox jump.

- **Role/Position:** Q3→Q4 wall (decompression/compression interface); the channel where energy “clicks” into mass is encoded by the depth ratio.

9.10 10. k_e (Coulomb Constant — EM Coupling)

- **Machine Value:** Emerges from $\alpha^{-1} = 137$ (EM coupling readout at the wall structure).

- **Observed Value:** $\approx 8.987551789 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ (inherits drift via α).

- **Derivation:**

- $\alpha^{-1} = 128 + 9 = 137$ (positional offset: op-span wall plus injection remainder).
- $\alpha = 1/137$.
- $k_e = \alpha \times (\hbar c/e^2)$ scaffold (falls directly from α in charge register).

- **Dimensional Threading:**

- 1D: $\alpha = 1/137$ baseline coupling.
- 2D: Threads through $\hbar c$ grounding.
- 3D: Drift in measured charge can be read as load in the charge register.
- 4D: Configures EM-gravity ratio via α/G .

- **Role/Position:** 90° wall in charge register; electromagnetic force scaling between charges.

9.11 Additional Notes on Related Constants

- **κ (Einstein Coupling; Observer Shadow):** Machine: $\Pi^{-8} = 1/512$ (exact dimensionless). Observed: $\kappa = 8\pi G/c^4$ (unit bridge). Role: Curvature-source lock in GR; why c^5 breaks: no fifth quadrant.

- **c (Propagation Limit; Bit-Rate Ceiling):** Machine: $c = u_1/u_2 = 128$ exactly. Observed: 299792458 m/s (speed of light). Role: Maximum info propagation; c^4 in κ as full circuit.

- 4π (**Four Quadrant Operations**): Machine: $4\pi_{\text{machine}} = 4\left(3 + \frac{9}{64}\right) = 12 + \frac{9}{16} = 12.5625$ exact. Observed: $\approx 12.566370\dots$ Role: Full sphere surface in machine register.

All constants tie back to the unity lock: $720^\circ = 1$ sphere, $360^\circ/360^\circ = 1$ operation.

10 Positions of the Constants in the NOS Gearbox Framework

Below is a compiled list of the positions for each constant from the provided NOS v11 framework constants list. Positions are specified in terms of angular coordinates (in degrees, NOSrad, or center-relative d), bit ranges (from the base lattice of 512 bits), quadrant/op assignments, ladder rungs ($\sqrt{N}$ tags from $\sqrt{1}$ to $\sqrt{512}$), and specific structural roles.

10.1 1. π (Geometric Aperture Ratio)

- **Angular Position:** Emerges at the 90° wall ($\theta = 90^\circ$; $\theta_{\text{NOS}} = 0.5$ NOSrad).
- **Bit Position:** 64 bits (linear lattice register, the 90° wall) plus the 9-bit remainder block; wall-scale readout is $3 + 9/64$; binary termination at bit-6.
- **Quadrant/Op Position:** Central to all four quadrants (Q1–Q4, ops x/y/z/w); spans the four-quadrant operations around the center pivot (d=0).
- **Ladder Rung:** $\sqrt{8}$ (90° wall, 8 octants).
- **Structural Role:** Ratio at the 90° wall; hull (three 64 bit wall-blocks) + active curvature zone (nine base steps); hexagon 6-fold collapse through center pivot.

10.2 2. α (Quadrant Coupling Ratio)

- **Angular Position:** At the 90° wall; alpha zone spans -180° to -167.34375° .
- **Bit Position:** $\alpha^{-1} = 137$ bits; alpha zone specifically 129–137 bits (9-bit injection), i.e., $137 = 128 + 9$.
- **Quadrant/Op Position:** Q1 (op x, charge/alpha quadrant); alpha zone in Q1 (post-Q2 saturation); dual-paired with Q3 (op z).
- **Ladder Rung:** $\sqrt{8}$ (90° wall).
- **Structural Role:** Sum readout built from op-span wall (128 bit) plus the nine-bit injection remainder.

10.3 3. G (Mass Compression Field)

- **Angular Position:** Deep in Q4 span ($d_{\text{deg}=180^\circ}$ to 360°); compression at the finest resolution near $\sqrt{512}$ rung.
- **Bit Position:** $1/128^4$ depth (u_4), tied to 512-bit closure.
- **Quadrant/Op Position:** Q4 (op w, mass/compression quadrant); dual-paired with Q2 (op y).

- **Ladder Rung:** $\sqrt{512}$ (base bit, finest 512 slices).
- **Structural Role:** Q4 mass/compression register; emerges as drift at inverse closure.

10.4 4. $\hbar$ (Threading Aperture)

- **Angular Position:** At hemisphere boundaries ($\theta = 360^\circ$; $\theta_{\text{NOS}} = 2 \text{ NOSrad}$).
- **Bit Position:** /256 bits (hemisphere partition from 512/2).
- **Quadrant/Op Position:** Between dual hemispheres (Q1/Q2 vs. Q3/Q4); aperture at center pivot (d=0).
- **Ladder Rung:** $\sqrt{2}$ (2 hemispheres).
- **Structural Role:** Action quanta aperture between hemispheres.

10.5 5. T_{CMB} (Heat Flow Baseline)

- **Angular Position:** Entry to Q3 span ($d_{\text{deg}=0^\circ}$ to 180°).
- **Bit Position:** 128/47 ratio; tied to 128 bits per quadrant.
- **Quadrant/Op Position:** Q3 entry (op z, thermal quadrant); dual-paired with Q1 (op x).
- **Ladder Rung:** $\sqrt{4}$ (4 quadrants/ops).
- **Structural Role:** Ground state heat signature at Q3 boundary.

10.6 6. E (Complete Inverse Measurement; Energy)

- **Angular Position:** Within Q3 ($d_{\text{deg}=0^\circ}$ to 180°); decompression near Q3→Q4 boundary.
- **Bit Position:** $u_1/u_3 = 16384$.
- **Quadrant/Op Position:** Energy state in Q3 (op z); bridges to Q4 (op w).
- **Ladder Rung:** $\sqrt{64}$ to $\sqrt{128}$ (mid-depth slices).
- **Structural Role:** Complete inverse measurement across hemispheres; energy-mass bridge.

10.7 7. u_n (Inverse Quadrant Depths)

- **Angular Position:** Per quadrant span: Q1–Q4.
- **Bit Position:** $1/128^Q$ per quadrant; $N_Q = 128^Q$ units.
- **Quadrant/Op Position:** Core to Q1 (op x), Q2 (op y), Q3 (op z), Q4 (op w); dual pairs $x \leftrightarrow z$, $y \leftrightarrow w$.
- **Ladder Rung:** Full ladder from $\sqrt{1}$ to $\sqrt{512}$.
- **Structural Role:** Unit hierarchy basis; local unit steps in each quadrant.

10.8 8. k_B (Boltzmann Constant)

- **Angular Position:** At Q3 boundary ($d_{\text{deg}=0^\circ}$, center pivot).
- **Bit Position:** Unit count at Q3 entry ($1/128^3 = u_3$ depth).
- **Quadrant/Op Position:** Entry to Q3 (op z, thermal quadrant).
- **Ladder Rung:** $\sqrt{32}$ to $\sqrt{64}$ (thermal slices).
- **Structural Role:** Access point to thermal register at Q3 boundary.

10.9 9. $g_{3 \rightarrow 4}$ (Thermal Channel Gain)

- **Angular Position:** At Q3→Q4 wall ($d_{\text{deg}=180^\circ}$).
- **Bit Position:** Fixed step between u_3 and u_4 , ratio of 128.
- **Quadrant/Op Position:** Q3→Q4 boundary (ops $z \rightarrow w$).
- **Ladder Rung:** $\sqrt{128}$ to $\sqrt{256}$ (deep partition).
- **Structural Role:** Channel wall where energy clicks into mass.

10.10 10. k_e (Coulomb Constant)

- **Angular Position:** At 90° wall ($\theta = 90^\circ$).
- **Bit Position:** From $\alpha^{-1} = 137$ (offset at $128 + 9$); charge register at 64 bits.
- **Quadrant/Op Position:** Q2 (op y, charge quadrant, full saturation 0–128 bits); extends to Q1 alpha zone.
- **Ladder Rung:** $\sqrt{8}$ (90° wall).
- **Structural Role:** EM force scaling at 90° wall in charge register.